
KCET – 2020

Number of Questions: 60

Total Marks: 120

Physics

1. The value of acceleration due to gravity at a height of 10km from the surface of earth is x . At what depth inside the earth is the value of the acceleration due to gravity has the same value x ?
 - A. 5 km
 - B. 20 km
 - C. 10 km
 - D. 15 km

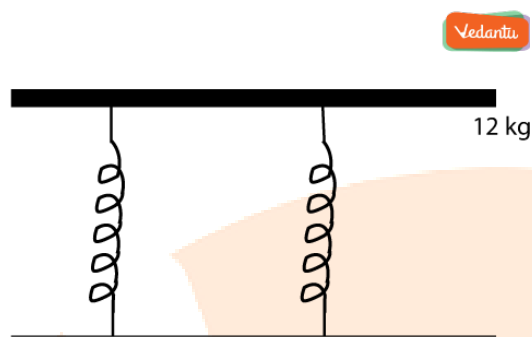
2. Young's modulus of a perfect rigid body is
 - A. Zero
 - B. Unity
 - C. Infinity
 - D. Between zero and unity

3. A wheel starting from rest gains an angular velocity of 10 rad/s after uniformly accelerated for 5 sec . The total angle through which it has turned is
 - A. 25 rad
 - B. 100 rad
 - C. $25 \pi \text{ rad}$
 - D. $50 \pi \text{ rad}$ about a vertical axis

4. Iceberg floats in water with part of it submerged. What is the fraction of the volume of iceberg submerged if the density of ice is $\rho_i = g \text{ cm}^{-3}$?
 - A. 0.917
 - B. 1
 - C. 0.458
 - D. 0

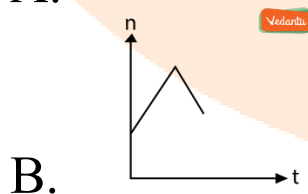
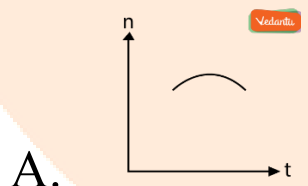
5. A sphere, a cube and a thin circular plate all of same material and same mass initially heated to same high temperature are allowed to cool down under similar conditions. Then the
 - A. plate will cool the fastest and cube the slowest
 - B. sphere will cool the fastest and cube the slowest
 - C. plate will cool the fastest and sphere the slowest
 - D. cube will cool the fastest and plate the slowest

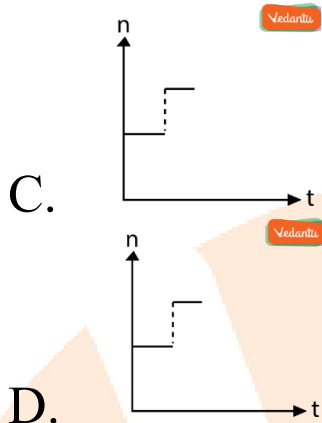
6. In an adiabatic expansion of an ideal gas the product of pressure and volume
- A. Decreases
 - B. Increases
 - C. Remains constant
 - D. At first increases and then decreases
7. A certain amount of heat energy is supplied to a monoatomic ideal gas which expands at constant pressure. What fraction of the heat energy is converted into work ?
- A. 1
 - B. $\frac{2}{3}$
 - C. $\frac{2}{5}$
 - D. $\frac{5}{7}$
8. A tray of mass 12 kg is supported by two identical springs as shown in figure. When the tray is pressed down slightly and then released, it executes SHM with a time period of 1.5s. The spring constant of each spring is



- A. 50Nm^{-1}
- B. 0
- C. 105Nm^{-1}
- D. ∞

9. A train whistling at constant frequency 'n' is moving towards a stationary observer on the station. The train goes past a stationary observer on the station. The frequency 'n' of the sound as heard by the observer is plotted as a function of time 't'. Identify the correct curve

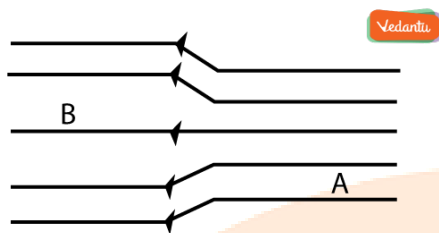




10. A point charge 'q' is placed at the corner of a cube of side 'a' as shown in the figure. What is the electric flux through the face ABCD ?

- A. 0
- B. $\frac{q}{24 \epsilon_0}$
- C. $\frac{q}{6 \epsilon_0}$
- D. $\frac{q}{72 \epsilon_0}$

11. The electric field lines on the left have twice the separation on those on the right as shown in figure. If the magnitude of the field at A is 40 Vm^{-1} , what is the force on $20 \mu\text{C}$ charge kept at B?



- A. $4 \times 10^{-4} \text{Vm}^{-1}$
- B. $8 \times 10^{-4} \text{Vm}^{-1}$
- C. $16 \times 10^{-4} \text{Vm}^{-1}$
- D. $1 \times 10^{-4} \text{Vm}^{-1}$

12. An infinitely long thin straight wire has uniform charge density of $\frac{1}{4} \times 10^{-2} \text{cm}^{-1}$. What is the magnitude of electric field at a distance 20 cm from the axis of the wire ?

- A. $1.12 \times 10^8 \text{NC}^{-1}$
- B. $4.5 \times 10^8 \text{NC}^{-1}$
- C. $2.25 \times 10^8 \text{NC}^{-1}$
- D. $9 \times 10^8 \text{NC}^{-1}$

13. A dipole moment 'P' and moment of inertia I is placed in a uniform electric field $\vec{E}$. If it is displaced slightly from its stable equilibrium position, the period of oscillation of dipole is

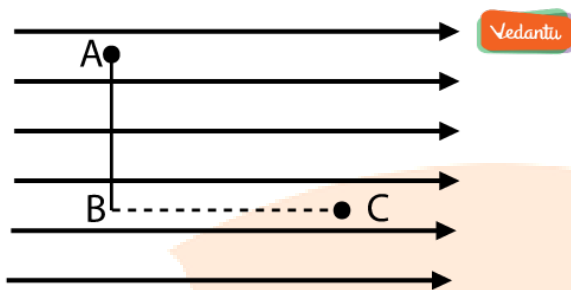
- A. $\sqrt{\frac{PE}{I}}$

B. $2\pi\sqrt{\frac{I}{PE}}$

C. $\frac{1}{2\pi}\sqrt{\frac{PE}{I}}$

D. $\pi\sqrt{\frac{I}{PE}}$

14. The difference between equivalent capacitances of two identical capacitors connected in parallel to that in series is $6\mu\text{F}$. The value of capacitance of each capacitor is
- A. $2\mu\text{F}$
 B. $3\mu\text{F}$
 C. $4\mu\text{F}$
 D. $6\mu\text{F}$
15. Figure shows three points A, B and C in a region of uniform electric field $\vec{E}$. The line AB is perpendicular and BC is parallel to the field lines. Then which of the following holds good? (V_A, V_B and V_C represent the electric potential at points A, B and C respectively)



- A. $V_A = V_B = V_C$
- B. $V_A = V_B > V_C$
- C. $V_A = V_B < V_C$
- D. $V_A > V_B = V_C$

16. When a soap bubble is charged ?

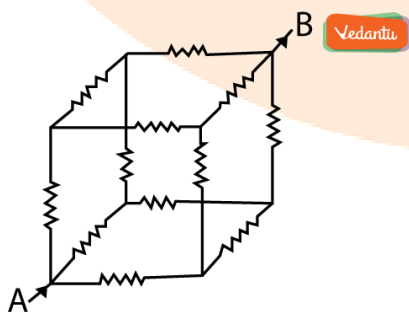
- A. Its radius increases
- B. Its radius decreases
- C. The radius remains the same
- D. Its radius may increase or decrease

17. A hot filament liberates an electron with zero initial velocity. The anode potential is 1200V. The speed of the electron when it strikes the anode is

- A. $1.5 \times 10^5 \text{ms}^{-1}$
- B. $2.5 \times 10^6 \text{ms}^{-1}$
- C. $2.1 \times 10^7 \text{ms}^{-1}$
- D. $2.5 \times 10^8 \text{ms}^{-1}$

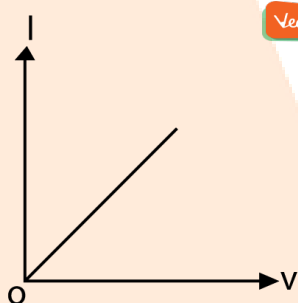
18. A metal rod of length 10 cm and a rectangular cross-section of $1\text{ cm} \times \frac{1}{2}\text{ cm}$ is connected to a battery across opposite faces. The resistance will be
- maximum when the battery is connected across $1\text{ cm} \times \frac{1}{2}\text{ cm}$ faces
 - maximum when the battery is connected across $10\text{ cm} \times \frac{1}{2}\text{ cm}$ faces
 - maximum when the battery is connected across $10\text{ cm} \times \frac{1}{2}\text{ cm}$ faces
 - same irrespective of the three faces
19. A car has a fresh storage battery of e.m.f 12V and internal resistance $2 \times 10^{-2}\Omega$. If the starter motor draws a current of 80A. Then the terminal voltage when the starter is on is
- 12V
 - 8.4V
 - 10.4V
 - 9.3V

20. A potentiometer has a uniform wire of length 5m. A battery of emf 10V and negligible internal resistance is connected between its ends. A secondary cell connected to the circuit gives balancing length at 200 cm. The emf of the secondary cell is
- 4V
 - 6V
 - 2V
 - 8V
21. The colour code for a carbon resistor of resistance $0.2\text{k}\Omega \pm 10\%$ is
- Red, Grey, Brown, Silver
 - Red, Green, Brown, Silver
 - Red, Grey, Silver, Silver
 - Red, Green, Silver
22. Each resistance in the given cubical network has resistance of 1Ω and equivalent resistance between A and B is



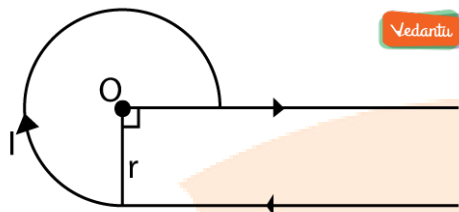
- A. $\frac{5}{6}\Omega$
- B. $\frac{6}{5}\Omega$
- C. $\frac{5}{12}\Omega$
- D. $\frac{12}{5}\Omega$

23. I-V characteristic of a copper wire of length L and area of cross-section A is shown in figure. The slope of the curve becomes



- A. More if experiment is performed at higher temperature
- B. More if a wire of steel of same dimension is used
- C. Less if the area of the wire is increased
- D. Less if the length of the wire is increased

24. In the given figure, the magnetic field at 'O'.

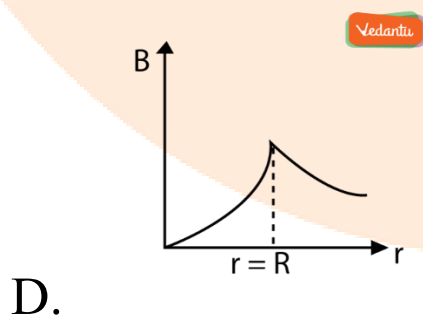
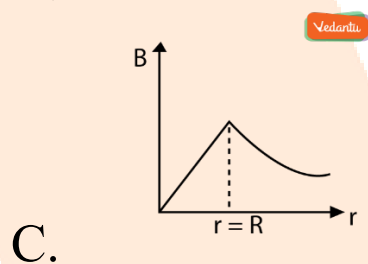
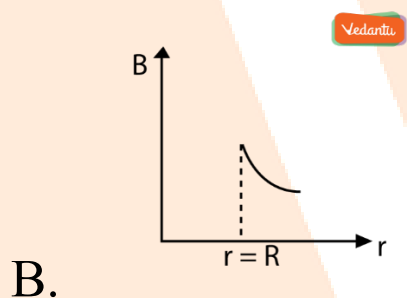
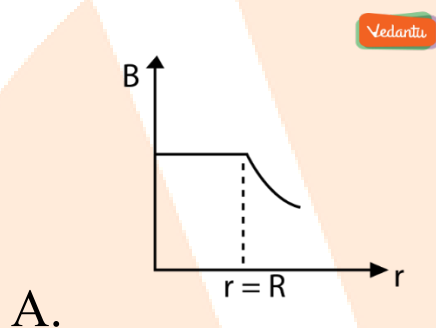


- A. $\frac{3}{4} \frac{\mu_0 I}{r} + \frac{\mu_0 I}{4\pi r}$
- B. $\frac{3}{10} \frac{\mu_0 I}{r} + \frac{\mu_0 I}{4\pi r}$
- C. $\frac{3}{8} \frac{\mu_0 I}{r} + \frac{\mu_0 I}{4\pi r}$
- D. $\frac{3}{8} \frac{\mu_0 I}{r} + \frac{\mu_0 I}{4\pi r}$

25. The magnetic field at the origin due to a current element $i \vec{dl}$ placed at a point with vector position $\vec{r}$ is

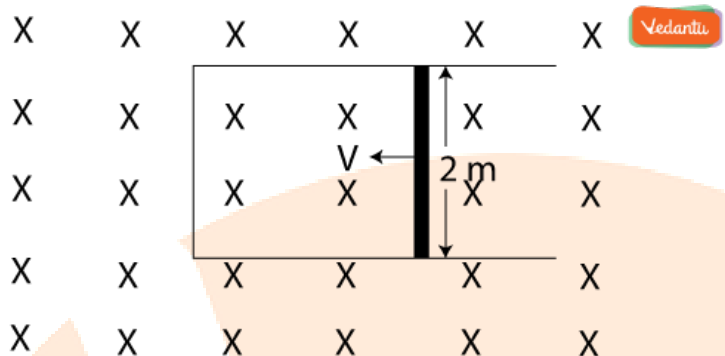
- A. $\frac{\mu_0 i}{4\pi} \frac{\vec{dl} \times \vec{r}}{r^3}$
- B. $\frac{\mu_0 i}{4\pi} \frac{\vec{r} \times \vec{dl}}{r^3}$
- C. $\frac{\mu_0 i}{4\pi} \frac{\vec{dl} \times \vec{r}}{r^2}$
- D. $\frac{\mu_0 i}{4\pi} \frac{\vec{r} \times \vec{dl}}{r^2}$

26. A long cylindrical wire of radius R carries a uniform current I flowing through it. The variation of magnetic field with distance ' r ' from the axis of the wire is shown by

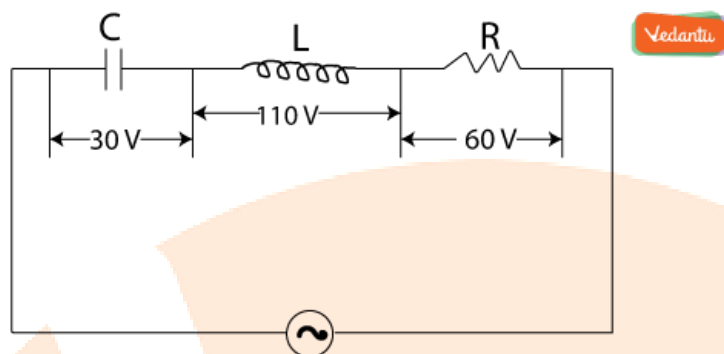


27. A cyclotron is used to accelerate protons (${}^1_1\text{H}$), Deuterons (${}^2_1\text{H}$) and α -particles (${}^4_2\text{He}$). While exiting under similar conditions, the minimum K.E. is gained by
- α -particle
 - Proton
 - Deuteron
 - Same for all
28. A paramagnetic sample shows a net magnetization of 8Am^{-1} when placed in an external magnetic field of 0.6T at a temperature of 4K . When the same sample is placed in an external magnetic field of 0.2T at a temperature of 16K . the magnetization will be
- $\frac{32}{3}\text{Am}^{-1}$
 - $\frac{2}{3}\text{Am}^{-1}$
 - 6Am^{-1}
 - 2.4Am^{-1}

29. The ratio of magnetic field at the centre of a current carrying circular coil to its magnetic moment is 'x' if the current and the radius both are doubled. The new ratio will become
- 2x
 - 4x
 - $\frac{x}{4}$
 - $\frac{x}{8}$
30. In a permanent magnet at room temperature
- Magnetic moment of each molecule is zero
 - The individual molecules have non-zero magnetic moment which are all perfectly aligned
 - Domains are partially aligned
 - Domains are all perfectly aligned
31. A rod of length 2 m slides with a speed of 5ms^{-1} on a rectangular conducting frame as shown in figure. There exists a uniform magnetic field of 0.04 T perpendicular to the plane of the figure. If the resistance of the rod is 3Ω . The current through the rod is



- A. 75 mA
B. 133 mA
C. 0.75 A
D. 1.33 A
32. The current in a coil of inductance 0.2 H changes from 5A to 2A in 0.5sec. The magnitude of the average induced emf in the coil is
A. 0.6 V
B. 1.2 V
C. 30 V
D. 0.3 V
33. In the given circuit the peak voltage across C, L and R are 30 V, 110 V and 60 V respectively. The rms value of the applied voltage is

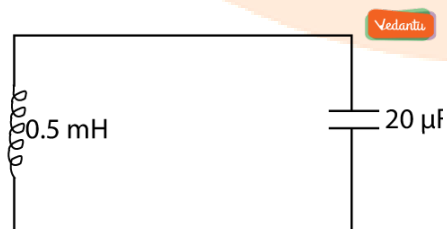


- A. 100 V
- B. 200 V
- C. 70.7 V
- D. 141 V

34. The power factor of R-L circuit is $\frac{1}{\sqrt{3}}$. If the inductive reactance is 2Ω . The value of resistance is

- A. 2Ω
- B. $\sqrt{2}\Omega$
- C. 0.5Ω
- D. $\frac{1}{\sqrt{2}}\Omega$

35. In the given circuit, the resonant frequency is

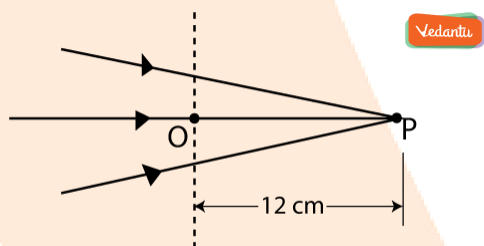


- A. 15.92 Hz
B. 159.2 Hz
C. 1592 Hz
D. 15910 Hz
36. A light beam of intensity 20W/cm^2 is incident normally on a perfectly reflecting surface of sides $25\text{cm} \times 15\text{cm}$. The momentum imparted to the surface by the light per second is
A. $2 \times 10^{-5} \text{kgms}^{-1}$
B. $1 \times 10^{-5} \text{kgms}^{-1}$
C. $5 \times 10^{-5} \text{kgms}^{-1}$
D. $1.2 \times 10^{-5} \text{kgms}^{-1}$
37. An object approaches a convergent lens from the left of the lens with a uniform speed 5 m/s and stops at the focus, the image
A. Moves away from the lens with an uniform speed 5 m/s
B. Moves away from the lens with an uniform acceleration
C. Moves away from the lens with a non- uniform acceleration
D. Moves towards the lens with a non-uniform acceleration

38. The refracting angle of prism is A and refractive index of material of prism is $\cot \frac{A}{2}$. The angle of minimum deviation is

- A. $180^\circ - 3A$
- B. $180^\circ + 2A$
- C. $90^\circ - A$
- D. $180^\circ - 2A$

39. The following figure shows a beam of light converging at point P. When a concave lens of focal length 16 cm is introduced in the path of the beam at a place shown by dotted line such that OP becomes the axis of the lens, the beam converges at a distance x from the lens. The value of x will be equal to



- A. 12 cm
- B. 24 cm
- C. 36 cm
- D. 48 cm

40. Three polaroid sheets P_1, P_2 and P_3 are kept parallel to each other such that the angle between pass axes of P_1 and P_2 is 45° and that between P_2 and P_3 is 45° . If unpolarised beam of light of intensity 128 Wm^{-2} is incident on P_1 . What is the intensity of light coming out of P_3 ?
- 128 Wm^{-2}
 - 0
 - 16 Wm^{-2}
 - 64 Wm^{-2}
41. Two poles are separated by a distance of 3.14 m. The resolving power of human eye is 1 minute of an arc. The maximum distance from which he can identify the two poles distinctly is
- 10.8 km
 - 5.4 km
 - 188 m
 - 376 m
42. In young's Double Slit Experiment, the distance between the slits and the screen is 1.2 m and the distance between the two slits is 2.4 mm. If a thin transparent mica sheet of thickness $1 \mu\text{m}$ and R.I. 1.5

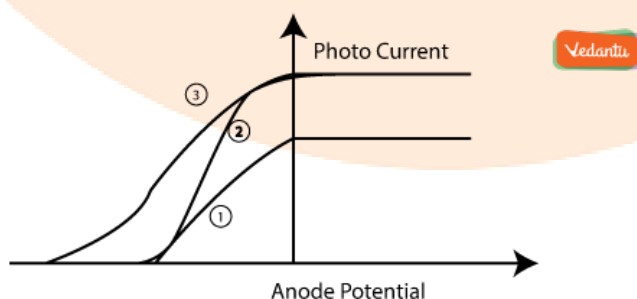
is introduced between one of the interfering beams, the shift in the position of central bright fringe is

- A. 2 mm
- B. 0.5 mm
- C. 0.125 mm
- D. 0.25 mm

43. The de-Broglie wavelength associated with electron of hydrogen atom in this ground state is

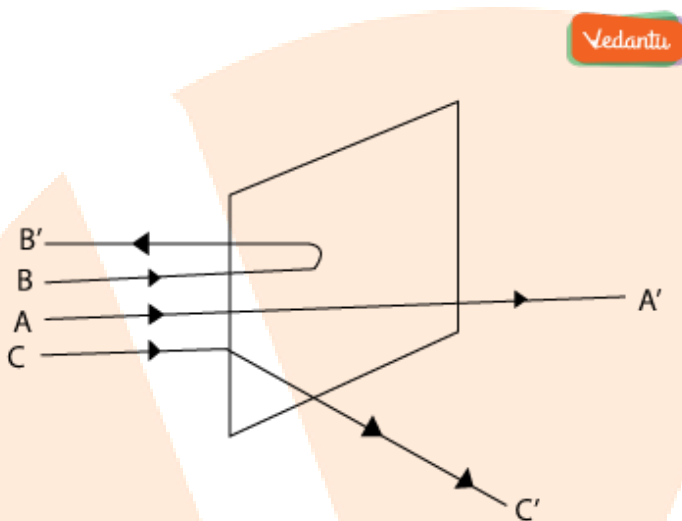
- A. $0.3 \text{ \AA}$
- B. $3.3 \text{ \AA}$
- C. $6.26 \text{ \AA}$
- D. $10 \text{ \AA}$

44. The following graph represents the variation of photo current with anode potential for a metal surface. Here I_1 , I_2 and I_3 represents intensities and γ_1 , γ_2 , γ_3 represent frequency for curves 1, 2 and 3 respectively, then



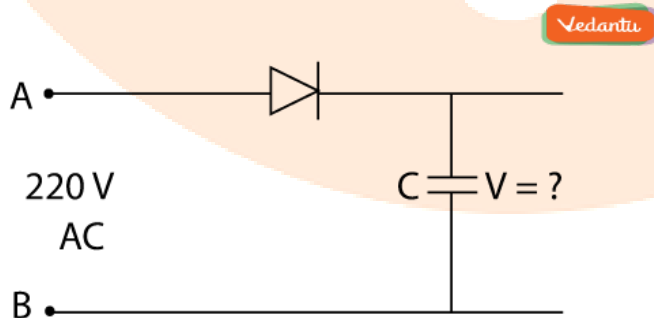
- A. $\gamma_1 = \gamma_2$ and $I_1 \neq I_2$
 B. $\gamma_1 = \gamma_3$ and $I_1 \neq I_3$
 C. $\gamma_1 = \gamma_2$ and $I_1 = I_2$
 D. $\gamma_2 = \gamma_3$ and $I_1 = I_3$
45. The period of revolution of an electron revolving in n th orbit of H-atom is proportional to
 A. n^2
 B. $\frac{1}{n}$
 C. n^3
 D. Independent of n
46. Angular momentum of an electron in hydrogen atom is $3h/2\pi$ (h is the Planck's constant). The K.E. of the electron is
 A. 4.35 eV
 B. 1.51 eV
 C. 3.4 eV
 D. 6.8 eV
47. A beam of fast moving alpha particles were directed towards a thin film of gold. The parts A, B and C of the transmitted and reflected beams corresponding to the incident parts A, B and C of the beam are shown

in the adjoining diagram. The number of alpha particles in



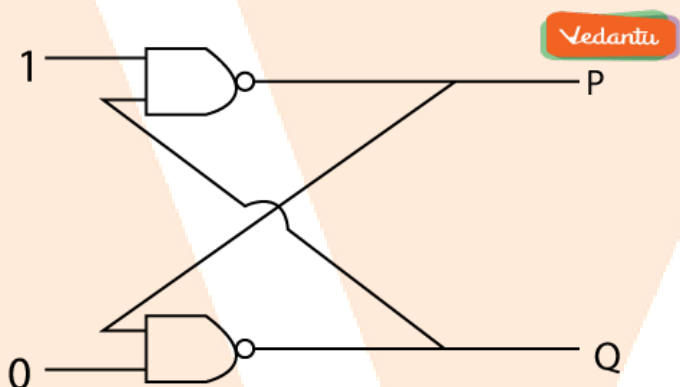
- A. B' will be minimum and in C' maximum
 - B. A' will be maximum and in C' minimum
 - C. A' will be minimum and in B' maximum
 - D. C' will be minimum and in B' maximum
48. Two protons are kept at a separation of 10 nm. Let F_n and F_e the nuclear force and the electromagnetic force between them
- A. $F_e = F_n$
 - B. $F_e \gg F_n$
 - C. $F_e \ll F_n$
 - D. F_e and F_n differ only slightly

49. During a β^- decay
- An atomic electron is ejected
 - An electron which is already present within the nucleus is ejected
 - A neutron in the nucleus decays emitting an electron
 - A proton in the nucleus decays emitting an electron
50. A radio-active elements has half-life of 15 years. What is the fraction that will decay in 30 years?
- 0.25
 - 0.5
 - 0.75
 - 0.85
51. A 220 V A.C supply is connected between points A and B as shown in figure what will be the potential difference V across the capacitor?



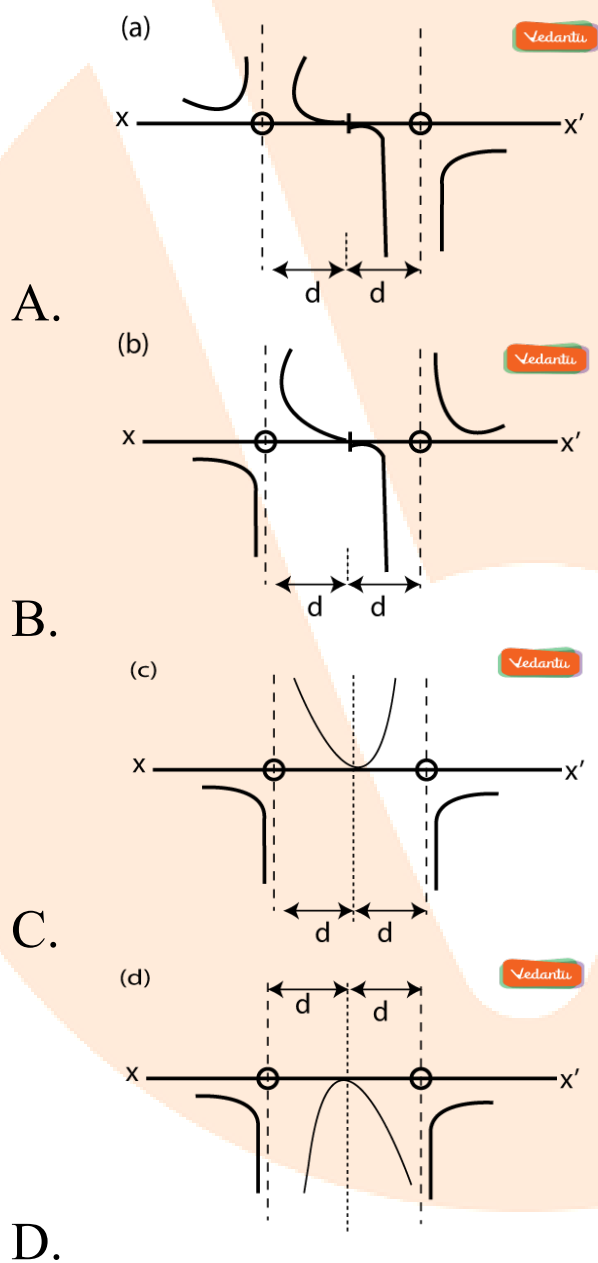
- A. 220 V
- B. 110 V
- C. 0
- D. $220\sqrt{2}V$

52. In the following circuit what are P and Q:



- A. $P = 1, Q = 0$
 - B. $P = 0, Q = 1$
 - C. $P = 0, Q = 0$
 - D. $P = 1, Q = 1$
53. A positive hole in a semiconductor is
- A. An anti-particle of electron
 - B. A vacancy created when an electron leaves a covalent bond
 - C. Absence of free electrons
 - D. An artificially created particle

54. Two long straight parallel wires are a distance $2d$ apart. They carry steady equal currents flowing out of the plane of the paper. The variation of magnetic field B along the line xx' is given by



55. A cylindrical wire has a mass $(0.3 \pm 0.003)\text{g}$, radius $(0.5 \pm 0.005)\text{mm}$ and length $(6 \pm 0.06)\text{cm}$. The maximum percentage error in the measurement of its density is
- 1
 - 2
 - 3
 - 4
56. At a metro station, a girl walks up a stationary escalator in 20 sec. If she remains stationary on the escalator, then the escalator take her up in 30 sec. The time taken by her to walk up on the moving escalator will be
- 25 sec
 - 60 sec
 - 12 sec
 - 10 sec
57. Rain is falling vertically with a speed of 12ms^{-1} . A woman rides a bicycles with a speed of 12ms^{-1} in east to west direction. What is the direction in which she should hold her umbrella?
- 30° towards East
 - 45° towards East

- C. 30° towards West
- D. 45° towards West

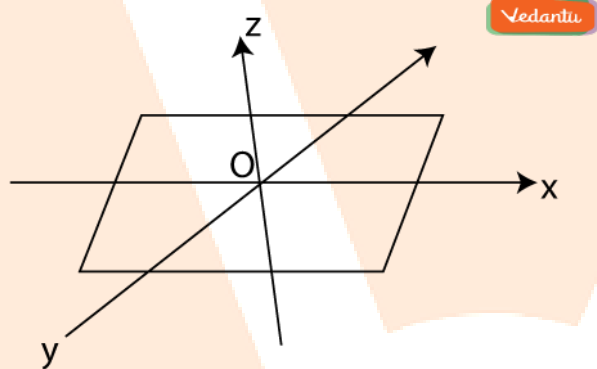
58. One end of a string of length 'l' is connected to a particle of mass 'm' and the other to a small peg on a smooth horizontal table. If the particle moves in a circle with speed 'v', the net force on the particle (directed towards the centre) is : (T is the tension in the string)

- A. T
- B. $T - \frac{mv^2}{l}$
- C. $T + \frac{mv^2}{l}$
- D. 0

59. A body is initially at rest. It undergoes one-dimensional motion with constant acceleration. The power delivered to it at time 't' is proportional to

- A. $t^{1/2}$
- B. t
- C. $t^{3/2}$
- D. t^2

60. A thin uniform rectangular plate of mass 2 kg is placed in X-Y plane as shown in figure. The moment of inertia about x-axis is $I_x = 0.2 \text{ kgm}^2$ and the moment of inertia about y-axis is $I_y = 0.3 \text{ kgm}^2$. The radius of gyration of the plate about the axis passing through O and perpendicular to the plane of the plate is



- A. 50 cm
- B. 5 cm
- C. 38.7 cm
- D. 31.6 cm